

Self-Regulating Mood Ecosystem

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Abstract

Self-Regulating Mood Ecosystem is an interactive installation exploring how affect emerges through gut microbial dynamics. The work treats ecological properties—delay, nonlinearity, competition, and recovery—as perceptible experience, foregrounding the complexity of gut–mood dynamics over simplified narratives of direct causation. A visitor selects a food item from a fast-food-style kiosk menu; food is used as input because everyday dietary choices are strongly linked to microbial community structure—clusters, diversity, and balance—over time. That choice adds a trace to the system’s shared mood, which unfolds through sound and light. Across encounters, traces accumulate into a collective psychology that visitors can influence but not fully control. The work externalizes what typically remains internal: the slow, distributed regulation of mood as an ecological process that can be collectively sensed, influenced, and inhabited.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; • **Applied computing** → **Media arts**.

Keywords

microbiome, gut–brain axis, interactive installation, embodied interaction, biofeedback, data physicalization, sonification, tangible interfaces

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1 Introduction & Background

The gut microbiome is increasingly discussed as a key actor in mood regulation [10, 14], yet it is often framed through simplified stories of direct causation [23]. In lived experience, changes in affect rarely arrive on command, and in microbial ecologies, shifts in balance, diversity, and community structure unfold through delay, feedback, and competition [4, 19]. Self-Regulating Mood Ecosystem starts from this mismatch. It treats the microbiome as an internal ecology

to be encountered—one that can be influenced through everyday inputs, while remaining partially autonomous. Externalizing these dynamics into a shared audiovisual environment, the work builds an interaction model in which meaning accrues gradually—across many visitors, through trace accumulation. This approach draws on somaesthetic design, which prioritizes felt, bodily experience [11].

1.1 Motivation

Self-Regulating Mood Ecosystem evolved from 95% Happiness (2022) [1], where we translated lived experience of mood regulation into a physical installation while engaging research that connects the gut microbiome to cognitive and affective processes, including associations discussed in conditions such as depression, autism, and schizophrenia [7, 9, 18]. The project began from an embodied question: what does it mean to treat mood as something regulated through an internal ecosystem that operates largely outside conscious awareness? Diagnosed with panic disorder and depression, the artist observed symptoms worsen alongside irregular eating and frequent relocation. Over two years of systematic observation—tracking microbiome profiles across environments and comparing them with medical records [1]—a parallel shift emerged: as psychiatric symptoms improved, the gut microbiome became more balanced and diverse. The gut is often described as a “second brain” because it is a site where neuroactive compounds can be produced or modulated by microbes, and because a large proportion of the body’s serotonin is produced in the gastrointestinal tract [23, 24]. For Self-Regulating Mood Ecosystem, these ideas motivate a central interaction design problem: how can visitors encounter mood regulation as a time-lagged, cumulative ecology—one that resists the expectation of immediate, controllable response? Food is used as an interface because it is ordinary yet consequential—an everyday input that plausibly perturbs microbial balance, allowing individual choices to accumulate into a shared system state.

1.2 Microbial Mindscales

The project developed through multiple iterations, including 95% Happiness (2022), where the artist’s medical records and self-tracked microbiome observations became a lens for thinking about mood as an embodied, ecological process [1]. In 2024, an early version of Self-Regulating Mood Ecosystem (Microbial Mindscales, 2024) used voice-based interaction: visitors spoke into the system, and the work responded by generating a mood trajectory from that utterance [2, 3]. We later shifted to a touchscreen kiosk interface. Voice incurred operational costs, raised data-policy concerns (voice is personally identifying), and introduced technical latency that



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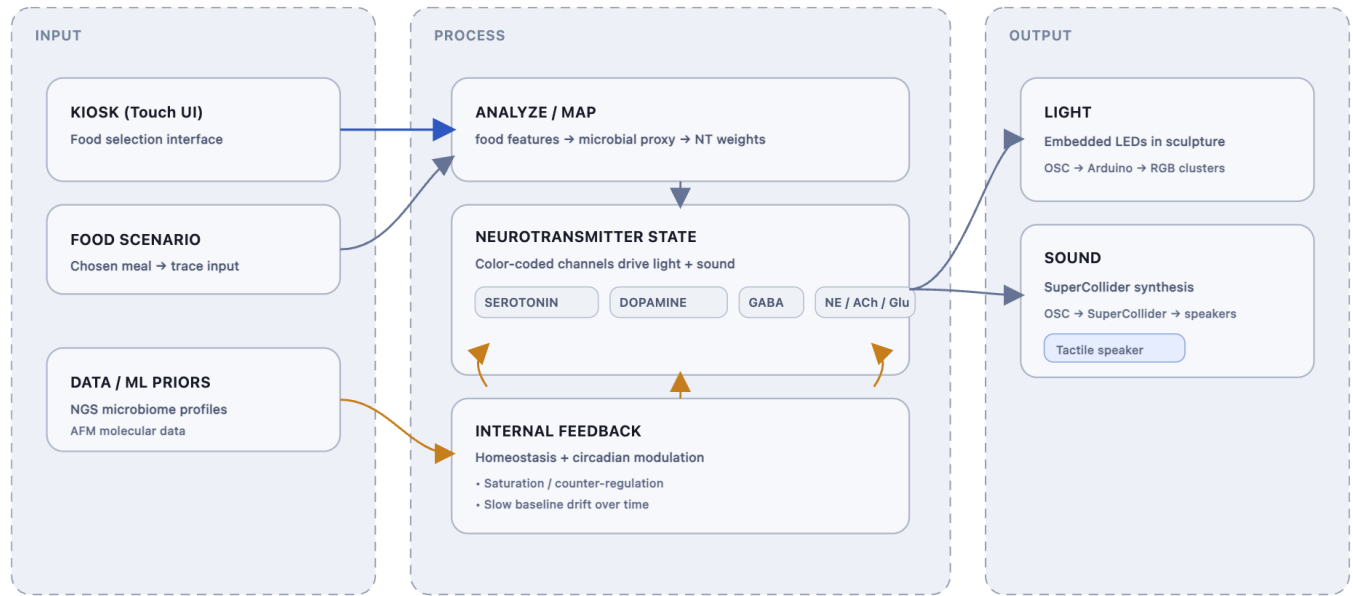


Figure 1: System architecture for Self-Regulating Mood Ecosystem. Food selections from the kiosk (left) are mapped through microbial and neurotransmitter models (center) to drive embedded LED and sound outputs (right). Internal feedback loops (orange arrows) maintain homeostasis and circadian drift when no visitor is interacting; visitor input (blue arrow) perturbs this baseline. The neurotransmitter state—informed by NGS microbiome profiles and AFM molecular data—serves as the bridge between food traces and audiovisual expression.

disrupted the experience—a delay that was technical rather than ecological.

The current demo reframes food selection as input to a creature-like sculptural body that digests it slowly. This framing draws on screen-based ordering as a familiar social ritual known to nudge people toward indulgent choices [21], treating food choice as shaped by environment and habit rather than purely individual decision. Each selection contributes a trace that accumulates across visitors, shaping an evolving shared mood.

1.3 Context and Contribution

Self-Regulating Mood Ecosystem draws on CHI work that treats interaction as embodied sense-making and that translates invisible processes into perceivable, material experience. In particular, research on data physicalization has argued that understanding can emerge through encounter—through form, temporality, and interpretation—rather than through a direct “readout” [5, 13]. It extends CHI work on first-person engagements with living matter and more-than-human perspectives, particularly in its framing of agency as distributed across visitors, system, and biological substrate rather than located in any single actor [6, 17]. Recent work on interoceptive interaction—designing for attention to internal bodily states—is also relevant [8]. Our contribution is a demoable interaction format for sensing gut–brain ecology as a collective, time-lagged process. Prior work has shown how interactive art installations can support participatory, cumulative experiences where each visitor’s input shapes what later participants encounter [12]; we extend this approach to biological data. Visitors do not “control” the work in a deterministic way; instead, each interaction adds a

trace that accumulates across participants, shaping an evolving shared mood that later visitors inherit. A demo format is essential because these dynamics are primarily temporal and experiential.

2 Interactive Experience

Self-Regulating Mood Ecosystem is experienced through a sculptural body that symbolizes the gut–brain axis (see Figure 1). Sound and light are embedded within the sculpture, so the system’s activity is perceived as internal—more like a living presence than an external display.

Step 1: Kiosk food selection (feeding the sculpture). A visitor selects a food item from a fast-food-style kiosk menu. Food is used because dietary choices are linked to microbial diversity and longer-term shifts in gut ecology. Selecting food is not treated as issuing a command; it is closer to feeding—offering nourishment, temptation, or habit to an externalized gut–brain ecology.

Step 2: Trace accumulation over exhibition time. After a selection, the choice is recorded as a trace in the work’s shared state. The system is not designed for instant feedback; a single selection produces only subtle change. What matters is accumulation: traces build up over the exhibition, and the sculpture’s mood shifts gradually. The system carries memory—later visitors inherit the altered state—so mood is experienced as collective rather than reset per person.

Step 3: Sculptural audiovisual unfolding driven by multi-source scientific data. The sculpture’s internal state is informed by three scientific datasets developed across the project. First, the artist’s gut microbiome data, obtained via next-generation sequencing (NGS), provides cluster and diversity profiles that determine

Neurotransmitter color coding + LED cluster mapping

Organoid-derived color analysis is encoded into a 6D neurotransmitter state, which bridges biological proxies and embedded light behavior. Hover a cluster to see its mapping.

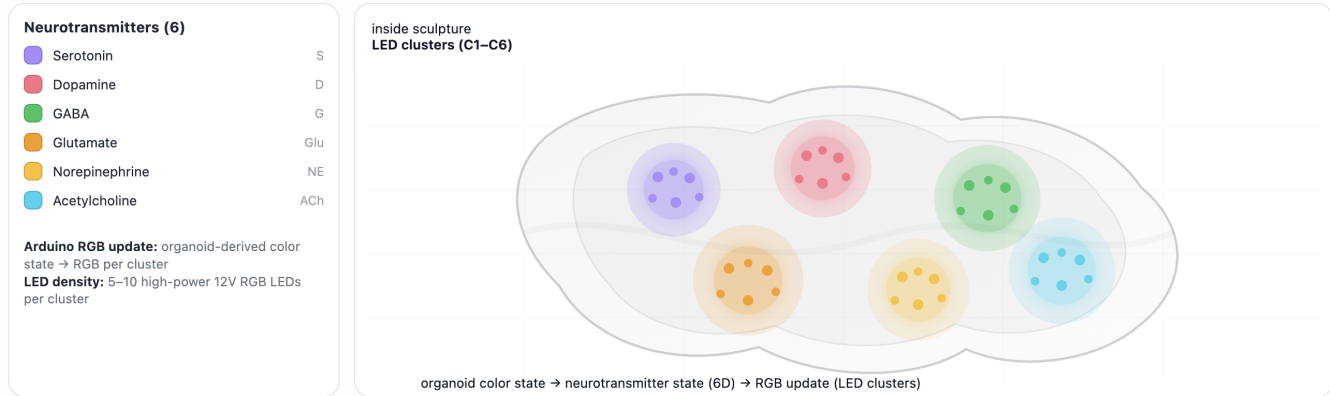


Figure 2: Neurotransmitter color coding and LED cluster mapping. Organoid-derived color analysis is encoded into a 6-dimensional neurotransmitter state, which serves as a bridge between biological proxies and the sculpture’s embedded light behavior. The Arduino updates each cluster’s RGB values in real time (5–10 high-power 12V RGB LEDs per cluster), producing localized color fields that drift as the shared state evolves.

how food traces relate to microbial community balance. Second, atomic force microscopy (AFM) data contributes molecular-scale measurements of neurotransmitter binding—specifically, dopamine binding to dopamine transporters and serotonin binding to serotonin transporters [25, 26]. These measurements shape the system’s representation of neurotransmitter dynamics without making literal diagnostic claims. Third, gut organoid imaging data, color-coded to show where serotonin- and dopamine-associated bacteria appear within the organoid structure [22], informs the spatial logic of the LED patterns.

Visitors perceive the system through embedded LEDs and sound inside the translucent body. Light appears as an internal glow with breathing-like modulation; sound conveys finer temporal change through shifts in density, roughness, and rhythm. Together, these outputs externalize internal regulation as a shared sensory ecology—one that can be influenced through feeding, but not fully controlled.

3 System Architecture & Implementation

This section describes how the experiential design outlined above is realized technically.

Self-Regulating Mood Ecosystem is implemented as a pipeline connecting kiosk input to a persistent internal state, which continuously drives the sculpture’s embedded light and sound (see Figure 1). The kiosk resets after each interaction, but the shared state does not; traces accumulate over exhibition time, producing gradual shifts rather than instant per-user results.

3.1 Hardware

The demo centers on a gut–brain-axis sculpture built from an SLA 3D-printed resin shell mounted on a metal frame, using organic geometry that houses outputs internally. An Arduino controls clusters of high-power 12V RGB LEDs (5–10 per microbiome cluster), updating RGB values in real time for localized glow, pulsing, and

breathing-like modulation (see Figure 2). Audio is played through a tactile speaker (surface transducer) attached to the sculpture; depending on site conditions, audio can also route to loudspeakers for multichannel diffusion. A touchscreen kiosk sits nearby for input, and a laptop coordinates light and sound.

3.2 Software

The kiosk runs an HTML-based menu interface that sends a minimal selection event (item ID and timestamp) to a persistent state model. This shared state accumulates traces over exhibition time and drives two coupled renderers: an Arduino-controlled LED engine that translates gut organoid mappings into cluster-based light patterns, and a SuperCollider [15] audio engine that sonifies the evolving state [16]. Sound is structured as an internal gut/brain feedback relationship—low-frequency (gut) and high-frequency (brain) layers modulate each other—and includes a timbral substrate from the artist’s stomach recordings; AFM molecular interaction data (dopamine/serotonin transporters and binding ligands) contributes additional fine-grained modulation that shapes micro-variations in the sound.

3.3 Interaction Logic and State Mapping

A food selection acts as a perturbation, not a direct trigger. Each menu item influences the virtual microbial ecosystem probabilistically: the same selection can produce different effects depending on current properties like diversity and stability. Over time, traces accumulate and shift the system’s position within an abstract space of brain-state “anchors” (e.g., anxiety, focus, lethargy), shaping audiovisual expression without labeling any individual visitor. Built-in stabilizers—saturation effects, diversity decay, delayed counterforces—prevent the system from getting stuck.

For light, microbiome clusters are expressed as weighted color fields; brightness reflects stability and transitions (see Figure 2). For

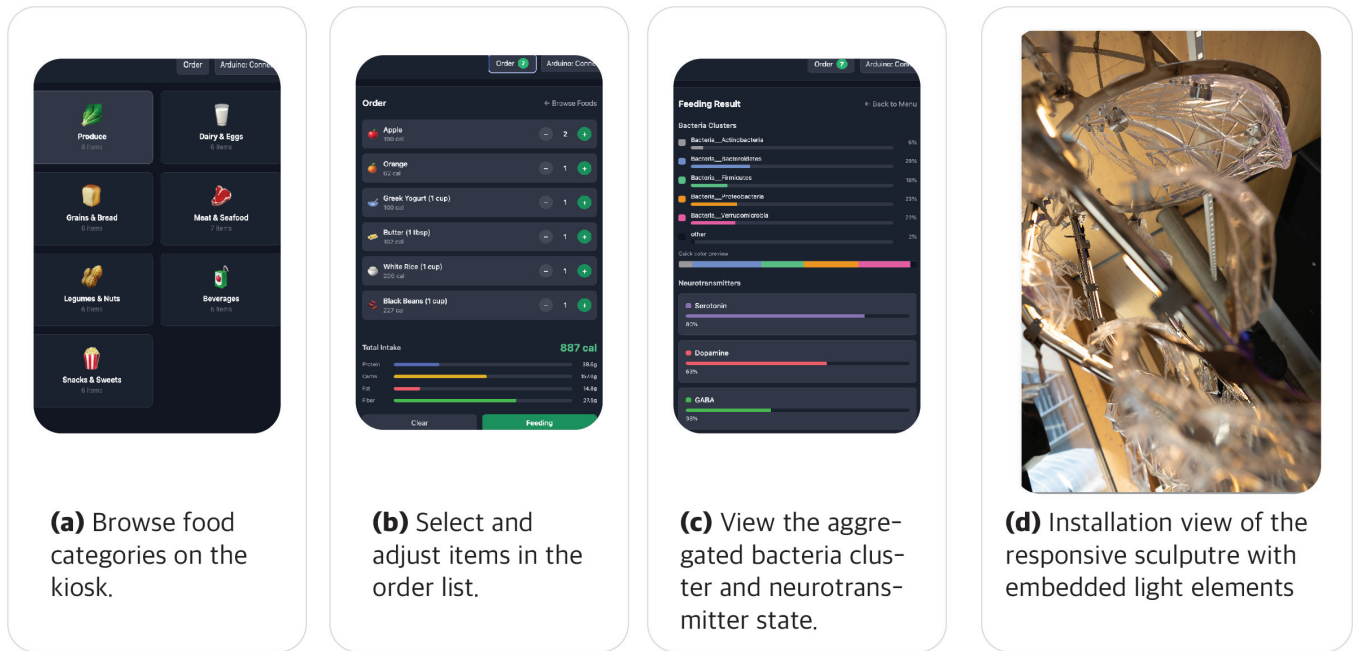


Figure 3: Kiosk interaction and embodied output in the demo setup. (a) Visitors browse food categories on the kiosk. (b) They select and adjust items to “feed” the system. (c) The interface summarizes the accumulating shared state as bacteria clusters and neurotransmitter levels. (d) The installation expresses this state through embedded light (LED clusters) and sound over time.

sound, the state drives low-frequency pulsation (gut motility) and high-frequency textures (cognitive activity), with limited mid-range so that response reads as drift rather than reaction. The system can also bias which kiosk items are visually emphasized, so each visitor inherits a choice environment shaped by prior input.

3.4 Setup Requirements

The installation can be floor-standing (approx. 1.7 m × 1.5 m) or tabletop. It runs offline, requiring standard AC power for kiosk, laptop, Arduino/LEDs, and audio. Multichannel routing is optional. Typical expo-hall conditions are acceptable; a small table for the kiosk and basic cable management are needed.

4 Conclusion

Self-Regulating Mood Ecosystem began as a question about what it means to encounter an invisible internal ecology rather than to visualize or measure it. Realizing the work clarified that the design challenge is temporal as much as material: building a system with memory, resistance, and drift—one that rewards return visits and collective use—is structurally different from designing per-user feedback loops. The hysteresis model proved generative here, and we expect to expand it in future iterations. By making accumulation, delay, and partial autonomy perceptible, the work offers an approach for experiencing invisible physiological processes as collective sense-making. The work treats scientific data not as a visualization target, but as material for artistic representation. We incorporate hysteresis (path-dependent memory) [20] to make

gut–brain regulation perceptible as a slow, stateful ecology. By translating microbiome profiles, organoid mappings, and molecular measurements into an evolving sensory environment, the work reframes gut–brain science as something that can be encountered and discussed in public.

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